Benchmarking Machine Learning Architectures for Speech Emotion Recognition

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***Abstract*—Speech Emotion Recognition (SER) is an emerging field that aims to automatically identify human emotions from spoken language to assist daily life. However, speech emotion recognition remains a challenging task due to the complex and dynamic nature of emotions. This study systematically benchmarks and evaluates a range of machine learning and deep learning architectures for SER, systematically evaluating their strengths and weaknesses under different conditions to provide clearer insights into which approaches are most effective, efficient, and stable. We found that while traditional Support Vector Machines (SVMs), particularly with HuBERT features, offer an excellent balance of high accuracy and superior efficiency, deep learning CNN-BiLSTM models provide competitive performance with WavLM features showing a slight edge in deep learning contexts. The composite efficiency metric proved instrumental in quantifying the trade-offs, demonstrating that highly efficient solutions can achieve strong predictive capabilities. The exploration of trending Large Acoustic Models within modular system designs has provided valuable insights for informed decision-making in developing effective and resource-aware SER applications.**

***Index Terms*—Speech emotion recognition, Acoustic feature extraction, Support vector machines, Convolutional neural networks, Hubert and wavlm representations**

I. Introduction

Speech Emotion Recognition (SER) is a field in computer science, the main objective of which is automatic recognition of human emotions from recorded spoken language.

It has applications in a wide variety of fields including education, healthcare, customer service. Nowadays, SER is one of the areas within artificial intelligence, as a big number of voice-controlled consumer devices have been engineered in the last decade, and a number of digital services, where it is important to recognize real people's emotions, have been introduced.

Despite some progress in this area, SER algorithms are far from being completely accurate and highly efficient. This research aims to benchmark machine learning architectures for SER and tries to develop one with a better balance of high accuracy and reasonable computational efficiency.

A. Gap Analysis

Many existing studies in speech emotion recognition focus mainly on getting the best accuracy, but not many look at the balance between how well the system works and how efficiently it uses resources. Many top performing models use big structures and lots of computing power, which makes it hard to use them in real world situations like on phones, in call centers, or in small devices. So, there's a big difference between models made for research that focus on being accurate and models that work well in daily life situations. This gap shows the need for a more balanced evaluation approach that considers both performance and efficiency.

B. Problem Statement

The main objective is to systematically benchmark and evaluate a range of machine learning and deep learning architectures for Speech Emotion Recognition (SER) in order to identify the optimal balance between model accuracy, generalization, and computational efficiency.

Specific objectives were the following:

1) to compare several promising architectures for SER in terms of performance and efficiency metrics;
2) to determine the composite metrics that capture the balance between performance and efficiency;
3) to explore and compare trending Large Acoustic Models such as Wave2vec2.0, HuBERT, and WavLM;
4) to explore modular and lightweight system design approaches; to establish a consistent evaluation framework that facilitates fair model comparison and supports informed decision-making for real-world SER applications.

C. Novelty of our work

The novelty of this study is paying special attention to efficiency metrics. Performance metrics are being calculated more frequently while evaluating ML architectures, and as a result of this study, a number of efficiency metrics were calculated and summarized.

D. Our Solutions

Within the study, several ML architectures for SER were implemented, and such characteristics of them as extraction and training time, MFLOPs, latency, efficiency score and number of parameters were calculated and analyzed.

II. Literature Review

Two approaches have been found out on the basis of recent reviews and studies conducted on speech emotion recognition. The first is the traditional method based on feature engineering, which relies on traditional machine learning classifiers such as Support Vector Machines (SVMs), Random Forests (RF), and K-Nearest Neighbors (KNN) classifiers in addition to traditional fixed feature extraction methods. Although traditional approaches have been widely known because of their ability to yield efficient results and interpretibility when employed on structured data, they have been found to be less capable at times in generalizing to other speakers and sets (Arun Kumar et al., 2025 [1]). But in recent studies, it has been found out that traditional approaches can achieve promising results if feature engineering is performed correctly. For example, on the research carried out by Rezapour Mashhadi & Osei-Bonsu (2023) [2], they had implemented a customized dataset which includes eight categories of emotions to determine the performance of the standard Random Forest Classifier and 1D CNN on the task. After the selection of features, the following manually engineered audio features of the dataset were extracted.

The second approach to emotion recognition is deep learning, which began receiving a considerable amount of attention beginning in 2019, and there was an increase in the amount of published research on this method in 2021 (Alhussein et al., 2025 [3]). Approaches such as Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory networks, and Gated Recurrent Units are found proven and have gained greater acceptance since these networks possess the ability of automatic learning of discriminative characteristics from the audio or its spectrogram (Arun Kumar et al., 2025 [1]). Also, Agrawal et al. (2023) [4] performed the analysis of various methods including traditional and Deep Learning methods such as SVM, k-NN, MLP, XGBoost, and CNN by using both RAVDESS and SAVEE emotion recognition datasets.

Notably, the year 2021 represented a tipping point in the use of hybrid feature extraction methods in the field of SER, given that 54% (7 of 13) of the studies reviewed appeared in that year compared to only 23% (3 of 13) that appeared prior to 2020 (Alhussein et al., 2025 [3]).

Particularly over the last few years, the SER field has registered immense progress with the use of Large Acoustic Models (LAM) such as Wave2vec2.0, HuBERT, Whisper, and WavLM. The success of these models is also contingent upon the use of the Transformer model, which has an exemplary capability for modeling sequential data such as audio (Alhussein et al., 2025 [3]). For example, from the paper presented by Iyer et al. (2022) [5], the authors proved that their Wav2Vec model outperformed the AlexNet CNN model with handcrafted features for three speech emotion datasets. Also, it can be noted that the most important element of the success of LAMs, resulting in their superiority for speech recognition tasks, is the concept of self-supervised learning.

Large acoustic models are typically trained upon large corpora of speech data. These corpora include thousands of hours of speech data recorded through various speakers and in different languages. Such acoustic models are also excellent for performing the concept of transfer learning. Several researchers (Wang et al., 2025 & Sun et al., 2024 [6]) initiated studying the concept of emotion analysis using the combination of a fine-tuned Wave 2Vec 2.0 acoustic model for the extraction of deep emotions and the utilization of deep learning algorithms for performing the classification of emotions.

It is pertinent to mention here that, in the study conducted by Wang et al. (2025) [6], a Neural Controlled Differential Equations (Neural CDEs) model was employed as a classifier. This model was able to generalize RNNs on the basis of controlled differential equations, which have been derived from rough path theory and stochastic analysis. It was observed that, during experimentation, not only did it surpass other baselines - with a WA of 73.37% and an UA of 74.18% - but it also demonstrated robustness with standard deviations of 0.45% for WA and 0.39% for UA. This may be because the model based on rough path theory considers the continuous evolution of the speech signal over time and the hidden space variable also changes with the input over time, making the model insensitive to the small perturbations commonly found in the data and enabling it to adapt more smoothly to different emotional expressions, which leads to a significant improvement in stability. Additionally, their model converged very quickly, reaching a good accuracy after just one epoch of training.

Another popular trend in the SER community is ensemble learning, which has been identified as a significant approach for enhancing the process of generalization. In a competition discussed by Naini et al. (2025) [7], the best-performing group attained excellent performance for categorical emotion recognition by using an ensemble of 14 individually trained fusion models. The team demonstrated that this approach utilizing multiple foundational models through deep hierarchical fusion was successful. Also, Chowdhury et al. (2025) [8] have identified an ensemble model that attained excellent performance by taking the average results of two base models, namely CNN and CNN-BiLSTM, for all five prominent publicly available datasets.

The multimodal approach is a consistent trend that enables richer representations for emotion recognition. The study of Naini et al. (2025) [7] demonstrated the complementary role of speech and text foundational models. Some works suggested models that utilize audio, video, and/or physiological signals as inputs to better recognize emotions. Despite the statistically significant robustness of unimodal speech (compared to multimodal)'s statistical significance for recognizing categorical emotional expression among participants in the modality metaanalyses of Al-Hussein, et al. (2025) [3], they limited their ability to reach certain conclusions regarding the combination of multiple data forms (video + physiological signals) into a model used for recognizing emotion. There is an opportunity for additional research to conduct multimodal recognition using multiformat data to better identify emotional signals.

III. Methodology

A. Dataset

The publicly available Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) was used [9]. It offers a balanced distribution of speakers, emotions, and intensity levels. Both classical and deep learning models can benefit from its clear emotion labeling and excellent audio recordings. The dataset contains 24 professional actors (12 female, 12 male), vocalizing two lexically-matched statements in a neutral North American accent. Speech includes calm, happy, sad, angry, fearful, surprise, and disgust expressions. The study applies a unimodal approach based solely on audio representations. Therefore, only 1,440 speech files will be considered as inputs (file example in Figure 1).

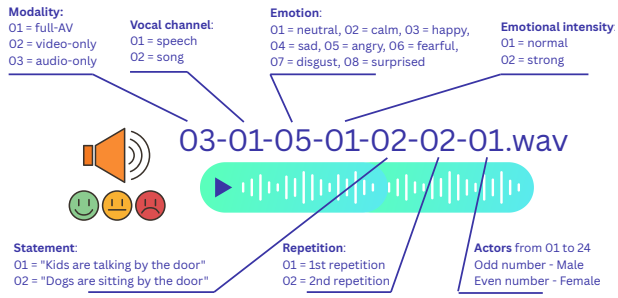

Fig. 1. Speech filename example.

B. Overall Workflow

The proposed experimental design aimed to evaluate the trade-off between model performance and efficiency across different architectures. For feature extraction, two LAMs, such as HuBERT and WavLM, were utilized due to their proven ability to capture high-level semantic and prosodic information from raw audio. For classification purposes, the Support Vector Machines (SVMs), Random Forest (RF), and the CNN-BiLSTM model were applied. Therefore, a comparison of a total of 6 architectures was made. Figure 2 represents the structured workflow of the performed experiment.

C. Experimental Settings

LAMs. The Large HuBERT self-supervised model was pretrained and fine-tuned on various emotion data (model name "superb/hubert-large-superb-er"). It is a task-specific model for Speech Emotion Recognition. The WavLM is a pretrained universal speech representation model based on the WavLM framework (model name "microsoft/wavlm-base-plus"). It is a modified HuBERT with WavLM enhancements that was trained on a diverse dataset (≈94k hours). The comparison of LAMs complexity is shown in Table I.

TABLE I
LAM MODELS COMPARISON.

Model	Layers	Hidden dimension	Parameters (Millions)	Model Size
Fine-tuned HuBERT	24	1024	≈315	≈1.2 GB
WavLM	12	768	≈94	≈360 MB

Both HuBERT and WavLM require 16 kHz raw speech waveform as inputs. During a feature extraction process, a middle layer's speech representations were used as output (8 level for HuBERT and 6 level for WavLM). The experiment showed that best SER accuracy achieved at intermediate layers.

Classifiers. After the feature extraction step, three different classifiers were utilized to predict the speaker's emotion. The comparison of classifiers' complexity is shown in Table II. The Support Vector Machines algorithm usually offers high accuracy compared to other classifiers. The main parameters used for this model were: linear kernel, regularization parameter $C = 1.0$, and gamma set to "scale". Another traditional classifier, Random Forest, was configured with 300 trees, using Gini impurity as the splitting criterion and balanced class weights.

TABLE II
CLASSIFIER MODELS COMPARISON.

Model	Parameters (Thousands)	Model Size
SVM (linear)	<29	≈6 MB
RF (300 trees)	<157	≈19 MB
CNN-BiLSTM (7 layers)	<972	≈4 MB

To ensure architectural diversity, the third classifier was selected from deep learning–based models. The CNN-BiLSTM model consisted of seven learnable layers and was trained using the Adam optimizer with a learning rate of $1 \times 10^{-}3$ and the Cross-Entropy loss function. The model was trained for five epochs.

Performance metrics. Classic performance metrics were used, including overall Accuracy, Precision, Recall, and F1 score. Before data splitting, the dataset was balanced across all emotion classes except neutral. However, after splitting into train and test sets, the class distribution became slightly imbalanced. As a result, the overall Accuracy corresponds to weighted accuracy (WA), as it depends on the class distribution. Precision, Recall, and F1 scores were computed using

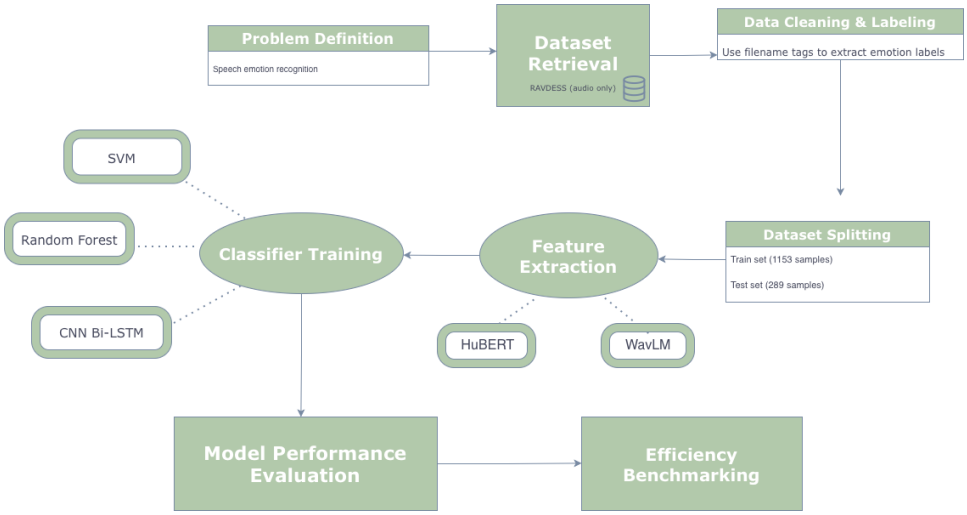

Fig. 2. Structured workflow diagram.

macro averaging, which assigns equal importance to each class.

Efficiency Metrics. Total training time (sec.), Training time per epoch (sec.), MFLOPs, and Latency (ms/sample) were calculated to assess the computational efficiency. FLOPs (Floating Point Operations) measure how many mathematical calculations a model performs to make one prediction. For the CNN-BiLSTM classifier, FLOPs were computed using the "thop" profiling library. The SVM and RF inference FLOPs per sample were estimated analytically based on Formulas (1) and (2), respectively. Fewer FLOPs mean that the prediction is faster and more efficient. Latency is the average time to predict one sample. Inference latency was measured as the average prediction time per sample over 1000 runs.

$$FLOPs = 2 \times d \tag{1}$$

where:
$d =$ number of input features per sample

$$FLOPs \approx N_{trees} \times D_{avg} \tag{2}$$

where:
$N_{trees} =$ number of trees
$D_{avg} =$ average depth of trees

Performance–efficiency tradeoff metric. The composite metric that captures the balance between performance and efficiency is presented by Formula (3). The proposed Efficiency score is not a widely-used metric, but it penalizes both large models and slow models, and scales well across classical ML and deep learning. It may be very useful for model comparison.

$$Efficiency\,Score = \frac{Accuracy(\%)}{Parameters(M) \times MFLOPs} \tag{3}$$

In this experiment, all computations were performed using Python (versions 3.12.3 and 3.12.12) with TensorFlow (versions 2.19.0 and 2.19.1) and PyTorch (versions 2.9.0+cpu and 2.9.1+cpu), implemented in Jupyter Notebook and Google Colab environments.

IV. Results

A number of efficiency and performance characteristics of different models and feature extractors was revealed in a result of benchmarking of various architectures for SER. Table III summarizes the performance metrics. Support Vector Machines (SVMs) which are using features from Large Acoustic Models (LAMs) demonstrated the highest F1 scores among traditional ML classifiers, specifically HuBERT+SVM achieved a macro F1 score of 0.9060, and WavLM+SVM achieved 0.8731. Random Forest (RF) models are also using LAM features, but their performance was lower. HuBERT+RF reached a macro F1 score of 0.6849, and WavLM+RF reached 0.6096. This is a result that indicates the advantage of linear SVMs.

Deep learning architectures demonstrated a competitive performance, but it was slightly varied. The macro F1 score achieved by HuBERT+CNN-BiLSTM model is 0.8658, in case of WavLM+CNN-BiLSTM model it was slightly higher and reached 0.8725. The bar charts comparing train and test performance for each model are presented in the Fig. 3 below. Their generalization capabilities and potential for overfitting can also be seen there. CNN-BiLSTMs demonstrated a good balance, and SVMs showed strong generalization.

From the efficiency point of view, the SVM models showed higher efficiency scores in terms of MFLOPs, latency, parameters. HuBERT+SVM and WavLM+SVM turned out to have significantly fewer parameters (0.022M for HuBERT+SVM and 0.029M for WavLM+SVM). These models also had lower MFLOPs compared to deep learning models (HuBERT+SVM had 2.36 and WavLM+SVM had 1.77). The CNN-BiLSTM models demanded higher computational resources, specifically HuBERT+CNN-BiLSTM had 0.97M parameters and 167.07 MFLOPs, and WavLM+CNN-BiLSTM had 0.81 parameters

TABLE III
PERFORMANCE METRICS

#	Architecture	Accuracy	Precision score	Recall score	F1 score
1	HuBERT+SVM	0.909722	0.906312	0.906594	0.905959
2	WavLM+SVM	0.881944	0.873754	0.874836	0.873067
3	HuBERT+RF	0.711806	0.708412	0.692378	0.684905
4	WavLM+RF	0.631944	0.623524	0.618799	0.609647
5	HuBERT+CNN-BiLSTM	0.895833	0.899683	0.882609	0.887895
6	WavLM+CNN-BiLSTM	0.850694	0.87058	0.849408	0.85057

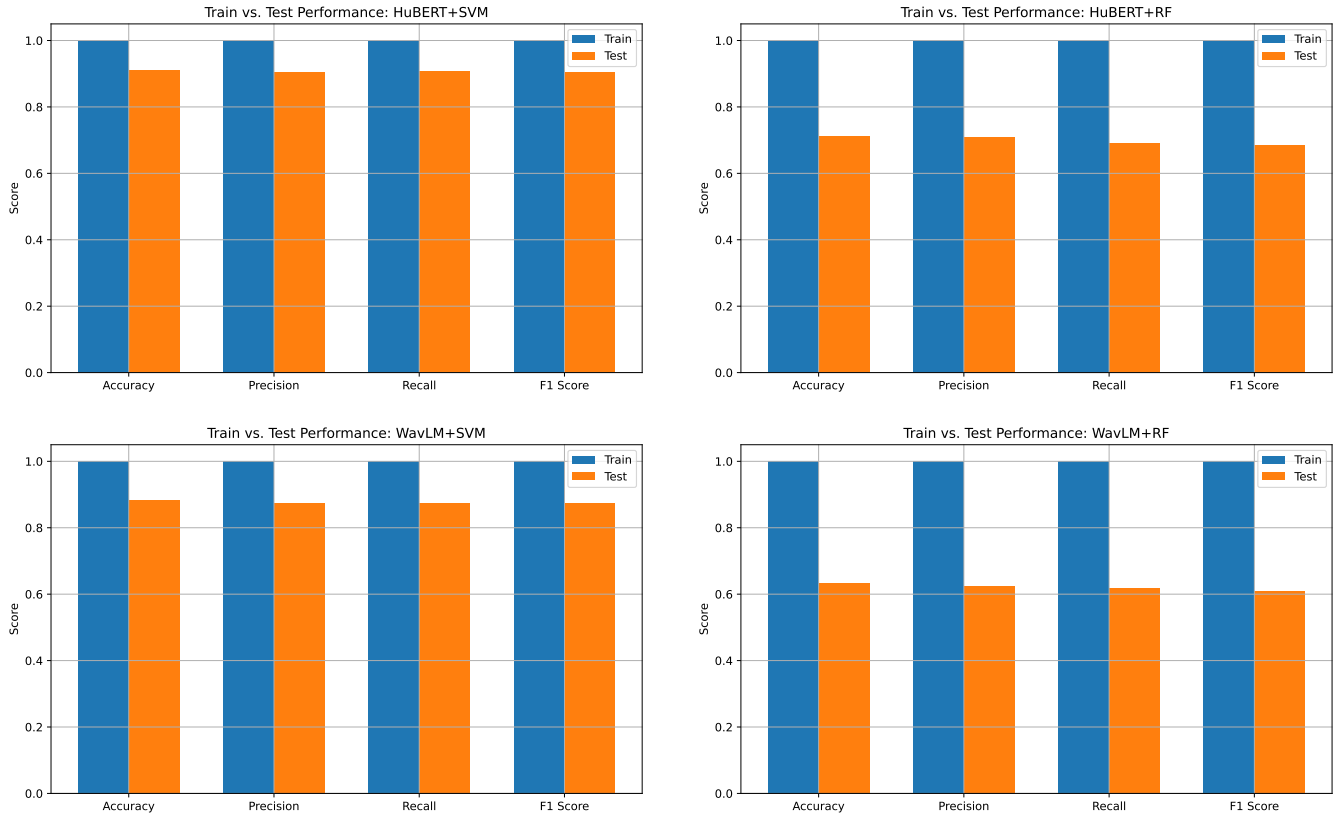

Fig. 3. Comparison of Train and Test Performance

and 128.08 MFLOPs. The detailed efficiency metrics are presented in Table IV.

Confusion matrices are presented in Fig. 4.

Training and validation parameters are illustrated as charts in Fig. 5.

Information of the system on which the architectures were implemented is the following:

- OS: Windows-11-10.0.26100-SP0
- CPU Model: AMD64 Family 25 Model 80 Stepping 0, AuthenticAMD
- CPU Cores (Physical): 6
- CPU Cores (Logical): 12
- RAM (Total): 15.34 GB
- RAM (Available): 1.95 GB
- GPU Name: No GPU detected

TABLE IV
EFFICIENCY METRICS

Efficiency Metrics of LAMs

#	Model	Extraction Time, sec
1	HuBERT	0.8017
2	WavLM	0.3389

Efficiency Metrics of Classifiers

#	Architecture	Training Time p/epoch, sec	Training Time Total, sec	MFLOPs	Latency (ms/sample)	Efficiency score	Parameters
1	HuBERT+SVM	N/A	0.197294	2.36	0.237445	1343.119943	28700
2	WavLM+SVM	N/A	0.151842	1.77	0.209215	2314.106661	21532
3	HuBERT+RF	N/A	1.100187	5.3	41.562219	94.60224199	141966
4	WavLM+RF	N/A	1.019323	5.45	41.591618	74.2194377	156230
5	HuBERT+CNN-BiLSTM	6.066148	30.330742	167.07	3.270354	0.551770392	971785
6	WavLM+CNN-BiLSTM	4.398086	21.990429	128.08	2.473332	0.822072751	807945

V. DISCUSSION

The linear SVMs, esperially when they are combined with HuBERT features, showed the highest F1 scores. This high perfomance, combined with low counts of parameters and MFLOPs, characterizes SVMs as highly efficient. Random Forest models did not achieve the lavel of perfomance of CNN-BiLSTMs or SVMs, which means, that in this case they were not as effective as the above mentioned models. Comparing the LAMs like HuBERT and WavLM demonstrated, that for traditional classifiers HuBERT was consistently reaching better results, but when integrated into the CNN-Bi-LSTM architecture, WavLM-based models showed a slight improvement of F1 score and better efficiency compared to HuBERT-based ones.

VI. CONCLUSION

The becnchmarking of machine learning and deep learning architectures for SER was successfully done. The efficiency metric, calculated from MFLOPs, parameters and accuracy, provides a quantitative measure of the blance between computational efficiency and the quality of results. It can be a valuable tool for informed decision making in cases, where this balance is an important factor.

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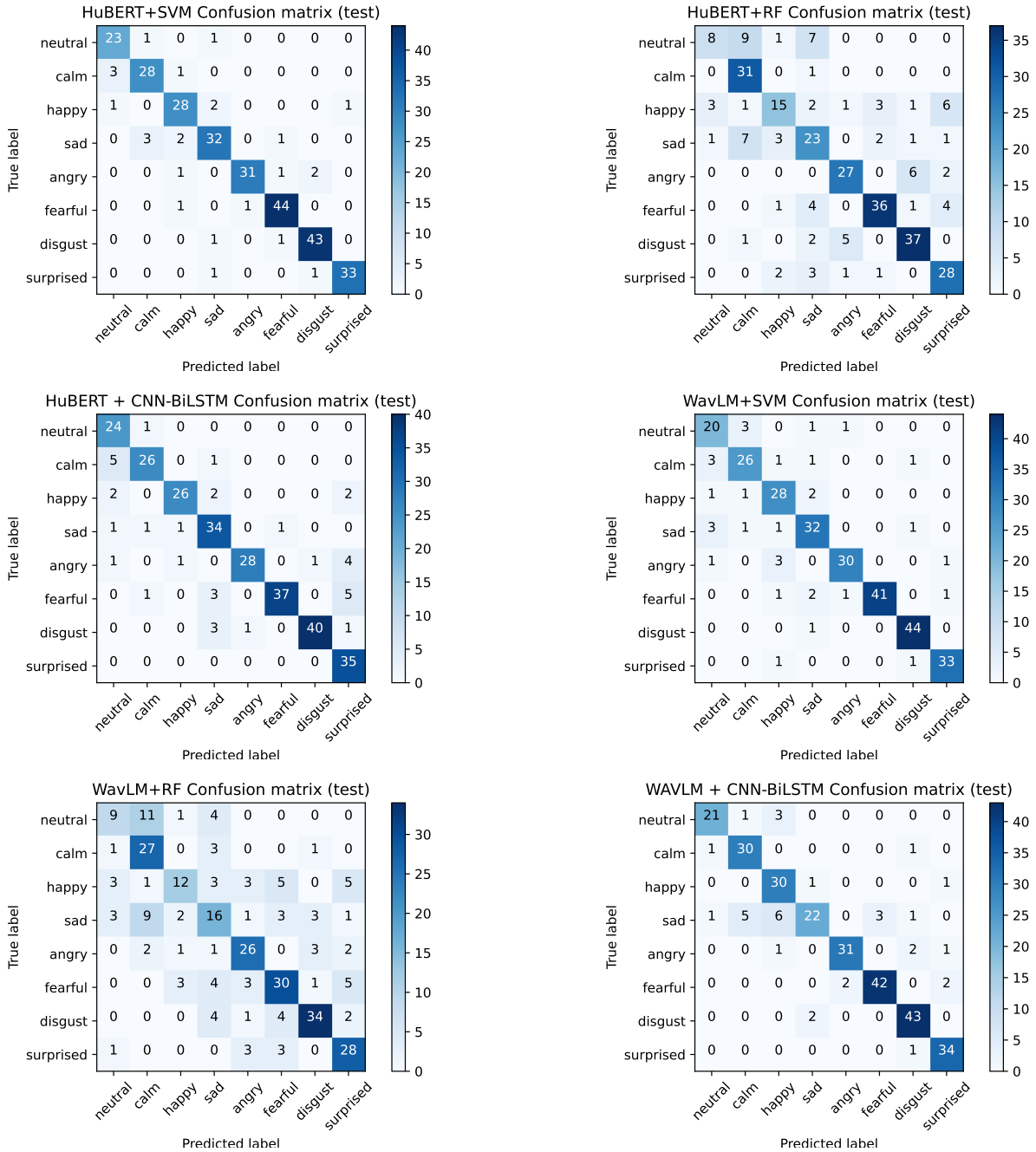

Fig. 4. Confusion matrices for different models.

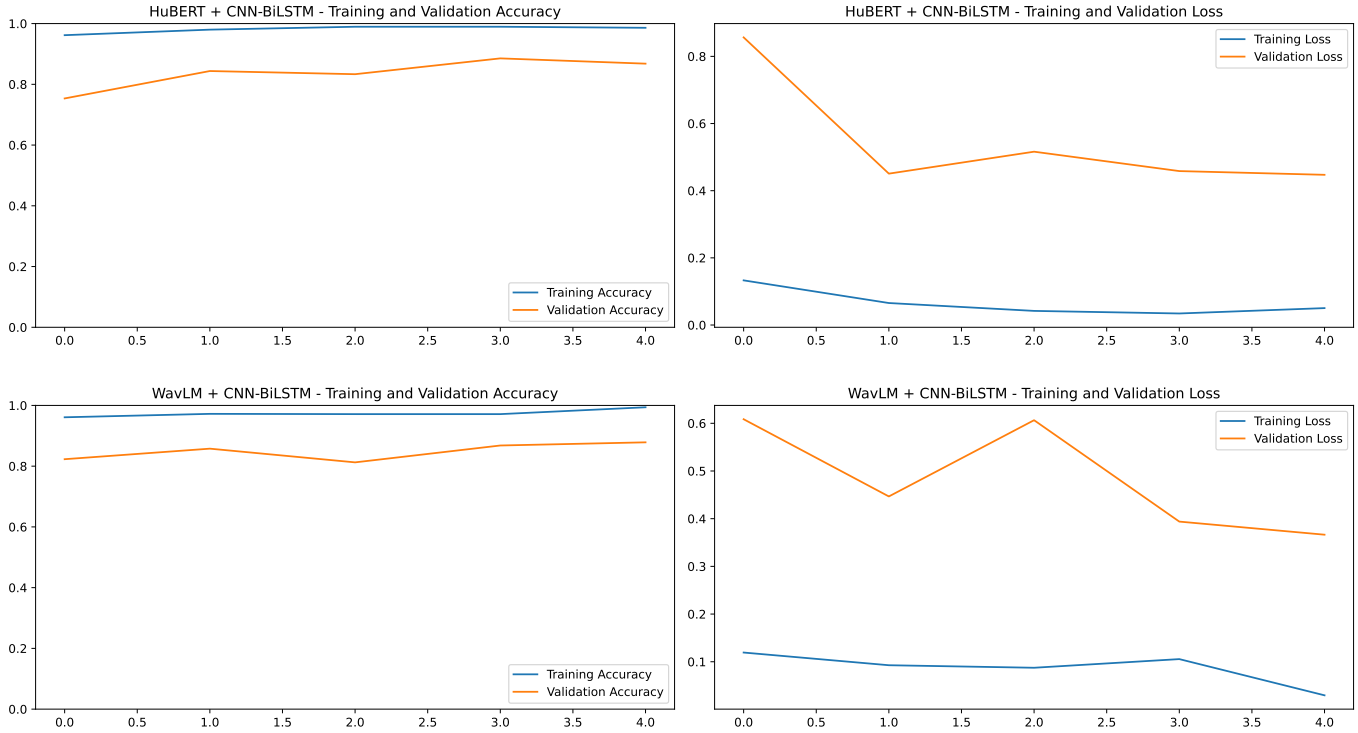

Fig. 5. Training and validation parameters